

Project Design Phase
Problem – Solution Fit Template

Date	01 October 2026
Team ID	SWTID-2026-4457
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- Solve complex problems in a way that fits the state of your customers.
- Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- Sharpen your communication and marketing strategy with the right triggers and messaging.
- Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- Understand the existing situation in order to improve it for your target group.

Template:

Define CS, fit into CC	1. CUSTOMER SEGMENT(S) CS <i>Who is your customer?</i> Primary: ServiceNow administrators, security admins and developers who must protect table data. Secondary: Academic institution staff (faculty, branch coordinators) managing student records. Learners: ServiceNow trainees practising ACL security in a 3-hour lab. End users: EEE branch users holding custom roles bb1-bb4.	6. CUSTOMER CONSTRAINTS CC <i>What limits their choice of solutions?</i> - Creating or editing ACLs needs the elevated security_admin role. - Needs scripting knowledge (gs.hasRole, return true/false). - Testing is only possible through user impersonation. - Limited lab time (3 hrs) and need for a ServiceNow instance. - Risk of locking out admins if the script is wrong.	5. AVAILABLE SOLUTIONS AS <i>What have they tried? Pros & cons?</i> Role-only ACLs: simple, but cannot check field values. Data-condition ACLs: easy, but limited logic. Business / query rules: flexible, but complex to maintain. UI policies / client scripts: hide data on screen, but not secure. Domain separation: powerful, but heavy for this use case.	Explore AS, differentiate
	2. JOBS-TO-BE-DONE / PROBLEMS J&P <i>Which jobs-to-be-done or problems do you address?</i> - Show only EEE-branch records to users with role bb1. - Control who can create (bb2), edit (bb3) and delete (bb4) records. - Keep full access for administrators. - Protect sensitive student data across forms, lists and Service Portal. Pain: data exposed to every user; over-permissive access.	9. PROBLEM ROOT CAUSE RC <i>What is the real reason this problem exists?</i> - Role-based ACLs alone cannot evaluate a record's field value (Branch). - Records of all branches sit in one shared table (u_institution_details). - Need for least-privilege access and data privacy. - Access is granted only when role, data condition and script all pass.	7. BEHAVIOUR BE <i>What does the customer do to get the job done?</i> - Creates user (EEE User) and roles bb1-bb4, then assigns them. - Builds the Institution Details table with ECE, EEE, CSE records. - Elevates to security_admin; creates Read, Create, Write, Delete ACLs. - Adds an Advanced script and data condition (Branch is EEE). - Impersonates users and admin to verify results.	
Focus on J&P, tap into BE, understand RC	3. TRIGGERS TR <i>What triggers customers to act?</i> - Sensitive records found visible to all users. - Need for branch-wise restriction or an audit/compliance request. - Unauthorized edit or delete of student records.	10. YOUR SOLUTION SL <i>Script-Controlled ACL on table u_institution_details</i> Read ACL (Advanced): role bb1, data condition <i>Branch is EEE</i>, plus a script. Script: return true for admin or bb1; otherwise return false (deny). Create ACL needs bb2; Write ACL needs bb3; Delete ACL needs bb4. Result: a record is accessible only when the ACL script evaluates to true. Verified by impersonation: bb1 user sees only EEE records; user without role sees none; admin sees all.	8. CHANNELS of BEHAVIOUR CH 8.1 ONLINE - What actions do customers take online? - Log in to ServiceNow as admin; elevate security_admin. - Use Application Navigator: Users, Roles, Tables, Access Control (ACL). - Impersonate users and open u_institution_details.list.	Extract online & offline CH of BE
	4. EMOTIONS: BEFORE / AFTER EM <i>How do customers feel before and after?</i> Before: worried, exposed, unsure who sees the data. After: confident, secure, in control of access.		8.2 OFFLINE CH <i>What actions do customers take offline?</i> - Follow the lab guide, documentation and screenshots. - Plan the role matrix (bb1-bb4); discuss with trainers or peers. - Review access policies and verification results.	

References:

- <https://www.ideahackers.network/problem-solution-fit-canvas/>
- <https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe>